

Spoken Tutorial – Python Essentials for AI

Title: Python Essentials for AI

Module	AI Essentials
Episode	Python Crash Course
Learning Objective	At the end of this tutorial learner will be able to 1. Use Google Colab environment 2. Create Python variables and strings 3. Construct lists and dictionaries 4. Define and execute functions
Approx. Duration	3-4 min
Outline	• Introduction to Python and AI • Setting up Google Colab • Variables and Strings • Working with Lists • Understanding Dictionaries • Creating Functions
Meta Tags	Python, AI, Google Colab, Programming, Coding Tutorial
Pre-requisite Tutorial	To follow this tutorial, learners need a Google Account and internet access.

Script

Visual Cue	Narration
Title slide with Python and AI logos	Welcome to this Spoken Tutorial on Python Essentials for AI. In this session, we will cover the absolute basics of Python needed for Artificial Intelligence.
List of learning objectives	In this tutorial, we will learn how to use Google Colab to write code. We will also create variables and strings. We will construct lists and dictionaries. Finally, we will define and execute functions.
Icons of web browsers and wifi signal	To record this tutorial, I am using the following setup. I am using a standard web browser like Chrome or Firefox. A stable internet connection is also required. Lastly, you will need a valid Google Account.
Checklist of prerequisites	To follow this tutorial, learners should have basic computer skills. You must also be logged into your Google Account. No prior installation of Python is required.

Python logo connecting to AI brain graphic	Let us start by understanding why we use Python. Quite simply, Python is the language of AI. It is famous for being easy to read. It almost looks like plain English. The global community has built incredible free tools for it.
Google Colab logo and browser interface	Next, let us look at our tool: Google Colab. Forget complex setups. We will use Google Colab because no installation is needed. It is a free tool from Google. It lets you write and run Python code directly in your web browser.
Screenshot of Google Colab welcome screen highlighting New Notebook button	Let us open a new notebook now. Step 1: Open your web browser. Step 2: Type <code>colab.research.google.com</code> in the address bar and press Enter. Step 3: Click the blue New Notebook button in the pop-up window. Wait for the notebook to load. You are now ready to write code.
Screenshot of Colab cell with variable code	Now, let us write our first piece of code. We will start with Variables and Strings. Think of a variable as a labeled box where you store information. A string is simply a piece of text wrapped in quotes. Step 1: Click on the code cell with the blinking cursor. Step 2: Type <code>'product_name = "Smart Coffee Mug"'</code>. Step 3: Click the Play button on the left side of the cell to run it. A green checkmark indicates it ran successfully.
Screenshot of Colab cell with list code	Next, let us move to Lists. A list holds multiple items in order, inside square brackets. Step 1: Hover your mouse below the previous cell and click + Code. Step 2: Type <code>'features = ["Temperature Control", "Wireless Charging"]'</code>. Step 3: Click the Play button to save the list. Now our variable features holds multiple items.
Screenshot of Colab cell with dictionary code	Let us now explore Dictionaries. A dictionary stores information in key-value pairs inside curly braces. It is perfect for organizing related data. Step 1: Create a new code cell. Step 2: Type the product details code as shown on the slide. This code defines the name and price. Step 3: Run the cell using the Play button. We have now structured our product data.

Screenshot of Colab cell with function definition and output	Finally, let us look at Functions. A function is a reusable block of code that performs a specific action. Step 1: Create a new code cell. Step 2: Type 'def create_greeting(name):' and press Enter. Notice the indentation automatically added. Step 3: Type 'return f"Welcome, {name}!"'. Step 4: Press Shift + Enter to define the function. Step 5: In a new line, type 'print(create_greeting("Alex"))' and run it. You should see the output: Welcome, Alex!
Summary slide with checkmarks next to topics	With this, we come to the end of this tutorial. Let us summarize. We learned how to use Google Colab without installation. We created variables to store text as strings. We built lists and dictionaries to organize data. And we defined functions to reuse code.
Assignment instructions text	As an assignment, please do the following in Google Colab. Step 1: Create a list of your top three hobbies. Step 2: Create a dictionary representing a pet. It should have keys for name and age. Step 3: Write a simple function that takes the pet's name and prints it. Verify that your code runs without errors.
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